Benjamin Wrinn

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EDUCATION

University of Pennsylvania - The School of Engineering and Applied Science — Philadelphia, PA

Candidate for BSE in Mechanical Engineering (3.25/4.0 UW) — *May 2029*

- **Courses:** Statics & Strength of Materials, Intro to Thermal Fluids, Linear Algebra, Calculus 1-3, Python

Oregon Episcopal School - High School — Portland, OR

- *Cumulative GPA:* 3.9/4.0 (UW); SAT: 1570 | Math 780 | Writing 790 — May 2025

RELEVANT EXPERIENCE

Vero Electric — Philadelphia, PA

Lead Mechanical Engineer — June 2026 – Present

- Engineered a custom, **261kWh/135kW** commercial peak-shaving **BESS enclosure** from SolidWorks to physical assembly, designing for -20 to 55°C ambient extremes and compliance with UL safety standards. (Certification by end of 2026)
- Optimized for low-cost manufacturing; validated architecture through SolidWorks **Thermal** and **Static** simulations
- Routed a 64-cell-series **BMS node PCB** in KiCad compatible with industry-leading battery manufacturers, cutting node integration time by enabling drop-in fit with existing pack architecture
- Now lead 2 engineers hired start of Fall; screened resumes and interviewed candidates

Penn Electric Racing Formula SAE EV Team — Philadelphia, PA

Mechanical Team — Sept 2025 – Present

- Designing **brake-absorption dynamometer** for team's new motor, testable with or without gearbox (500 N·m geared, 20,000 RPM direct); **improved torque measurement** to quantify gearbox losses.
- Engineering a servo-actuated Drag Reduction System projected to cut **~38%** of straight-line drag and **~2s** lap times
- Produced KeyShot renders used in sponsor and design-review presentations. Modeled and drew chassis components in SolidWorks per GD&T standards. Manufactured carbon fiber parts via resin infusion at sponsor facilities

American Rocketry Challenge — Portland, OR

Team Host, Hardware and Media Lead — Sept 2022 – August 2025

- Hardware lead on team's air brake system that autonomously adjusted drag via lookup table, landing within **7ft of target apogee** (**1st** in State, **20th** nationally, **Top 5** in Presentation)
- **Designed flight controller PCB**; ran CFD (SolidWorks) to predict drag before flight testing and populate lookup table
- Grew team Instagram by +5.5k followers and +350k interactions as Media Lead, driving sponsor visibility and recruitment

Children's Cancer Therapy Development Institute — Portland, OR

Student Research Assistant — June 2023 – June 2025

- Examined feasibility of quail embryo drug trials as manuscript co-author (pending publication); Wet-lab experience in both CAM and Bioluminescence assays
- Fabricated a prototype for machine vision on quail embryo drug trials, using OpenCV to quantify embryo health; cc-TDI has continued to develop this as an economical screening method to pair with mouse clinical trials.

ADDITIONAL

- **Skills: CAD** (SolidWorks – parts, assemblies, drawings, GD&T), **Manufacturing** (CNC Machining, 3D Printing, Laser Cutting, Wet Layups), **Simulations** (ANSYS, SolidWorks FEA, Luminary Cloud), **Kicad**/Altium PCB Design, **Python**, Blender, Davinci Resolve, Chinese (Mandarin), Completed *NFPA 70E* Training
- **Activities:** Weightlifting, USA National Kung Fu Team (Gold, Silver Medalist @ 2023 IWUF, Youth Coach), William Temple House Food Pantry Volunteer (stocked 3700+ lb of pantry items, fulfilled 120+ orders, coded menu translator in Python), Baking